**TITLE: Company Database**

# PART1: Requirements Analysis

The main operations of the company are production, communication with customers and project creation lead us to determine our entities according to what objects we need for a process to be done, also what attribute we need to describe every entity. The attribute depends on what data we need to store about every entity in our database

One Company can have many Managers but one Manager cannot work for many Companies.
Company are identified by Company ID, Company Name and Address.

One Manager can manage many Departments but one Department cannot have many Managers.
Manager are identified by Manager ID, Name and Phone number, Company ID and Salary.

One Department can have many Employees but one Employees can not belong to many Departments and One Department can have many products but one product can not belong to many Departments.
Department ID, Name, and Manager ID

One Employee can work on many Orders and also one Order can be worked on by many Employees.
Employee ID, Name, Phone number, Salary and Department ID.

One Product can be included in many Orders and also one Order can contain many Products.
Product ID, Name, Price and Department ID.

One Order cannot be placed to many Customers but one Customer can place many Orders.
Order ID, Employee ID, Product ID and Order date.

## PART 2: Conceptual Design:

The various steps before constructing ER Diagram are:

Step 1: Identify entities:

- **Company**
- **Employee**
- **Department**
- **Manager**
- **product**
- **Customer**
- **Order1**

- Product **Entities** and their **Attributes**
  - Company Entity: Attributes of Company Entity are Company ID, Company Name and Address.

    Company_ID is the primary Key for Company Entity.

  - Manager Entity: Attributes of Manager Entity are Manager ID, Name and Phone number, Company ID and Salary.

    M_ID is the primary Key for Manager Entity.

  - Department Entity: Attributes of Department Entity are Department ID, Name, and Manager ID.

    Dept_ID is the primary Key for Department Entity.

  - Employee Entity: Attributes of Employee Entity are Employee ID, Name, Phone number, Salary and Department ID.

    Emp_ID is the primary Key for Employee Entity.

  - Product Entity: Attributes of Product Entity are Product ID, Name, Price and Department ID.

    P_ID is the primary Key for Product Entity.

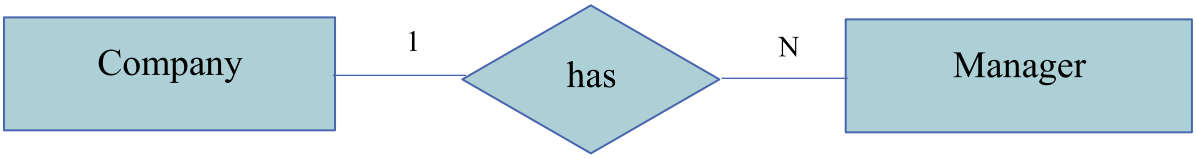
- Order1 Entity: Attributes of Order Entity are Order ID, Employee ID, Product ID and Order date.

  O_ID is the primary Key for Order1 Entity.

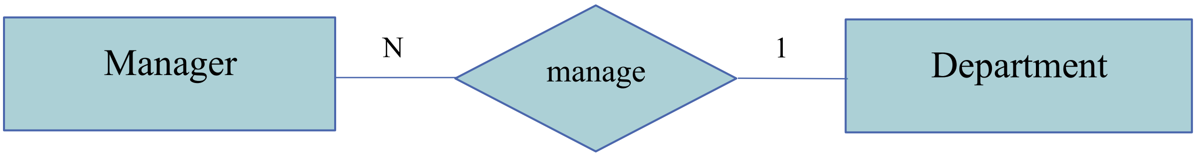
- Customer Entity: Attributes of Customer Entity are Customer ID, Name, Address, Email and Order ID.

  C_ID is the primary Key for Customer Entity.

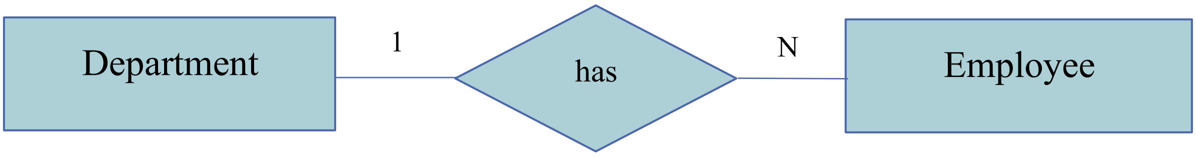
- **Relationships are:**
  - **Company has Managers** **(one to many)**
    One Company can have many Managers but one Manager can not work for many Companies, so the relationship between Company and Manager is one to many.




  - **Managers manages Department** **(many to one)**
    One Department can have many Managers but one Manager can not manage many departments, so the relationship between Manager and Department is many to one.

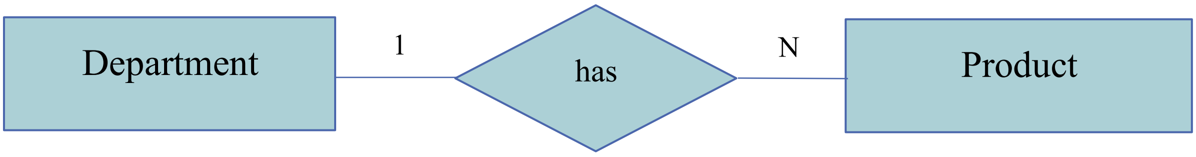



  - **Department has Employees** **(one to many)**
    One Department can have many Employees but one Employees can not belong to many Departments, so the relationship between Department and Employee is one to many.

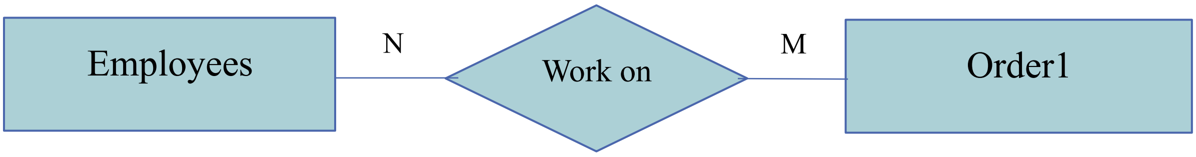



  - **Department has Products** **(one to many)**

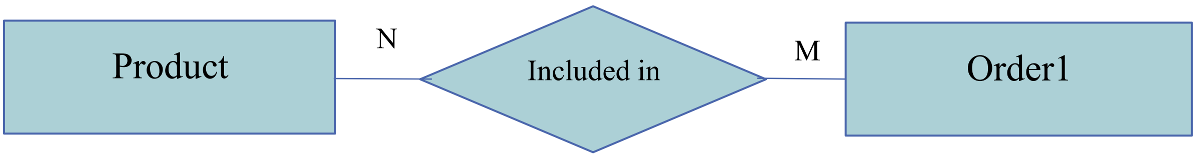
One Department can have many products but one product can not belong to many Departments, so the relationship between Department and product is one to many.



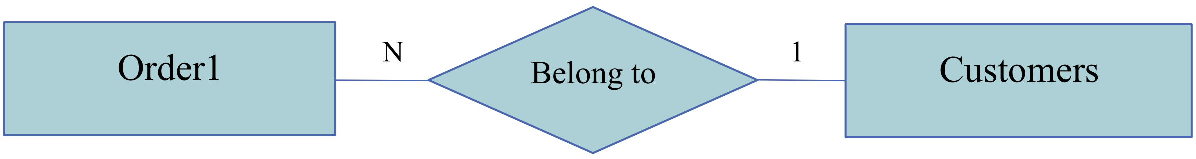

- **Employees work on Orders (many to many)**
  One Employee can work on many Orders and also one Order can be worked on by many Employees, so the relationship between Employee and Order are many to many.



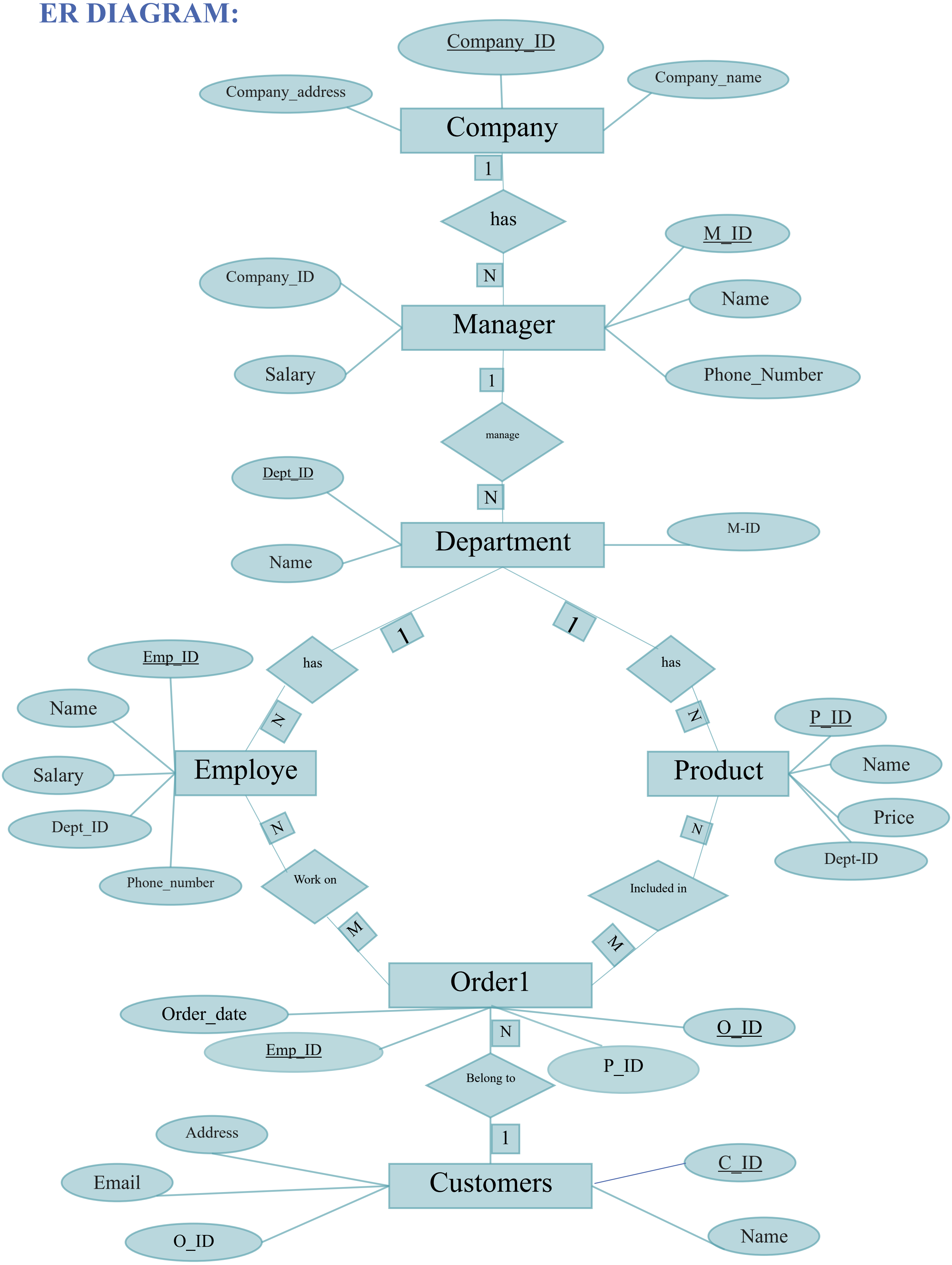

- **Products included in Orders (many to many)**
  One Product can be included in many Orders and also one Order can contain many Products, so the relationship between Product and Order are many to many.




- **Order belongs to customers (many to one)**
  One Order can not be placed to many Customers but one Customer can place many Orders, so the relationship between Order and Customers are many to one.



## ER DIAGRAM:


Company_ID
Company_address
Company_name
Company
1
has
N
M_ID
Company_ID
Name
Manager
Salary
Phone_Number
1
manage
N
Dept_ID
Department
M-ID
Name
1
1
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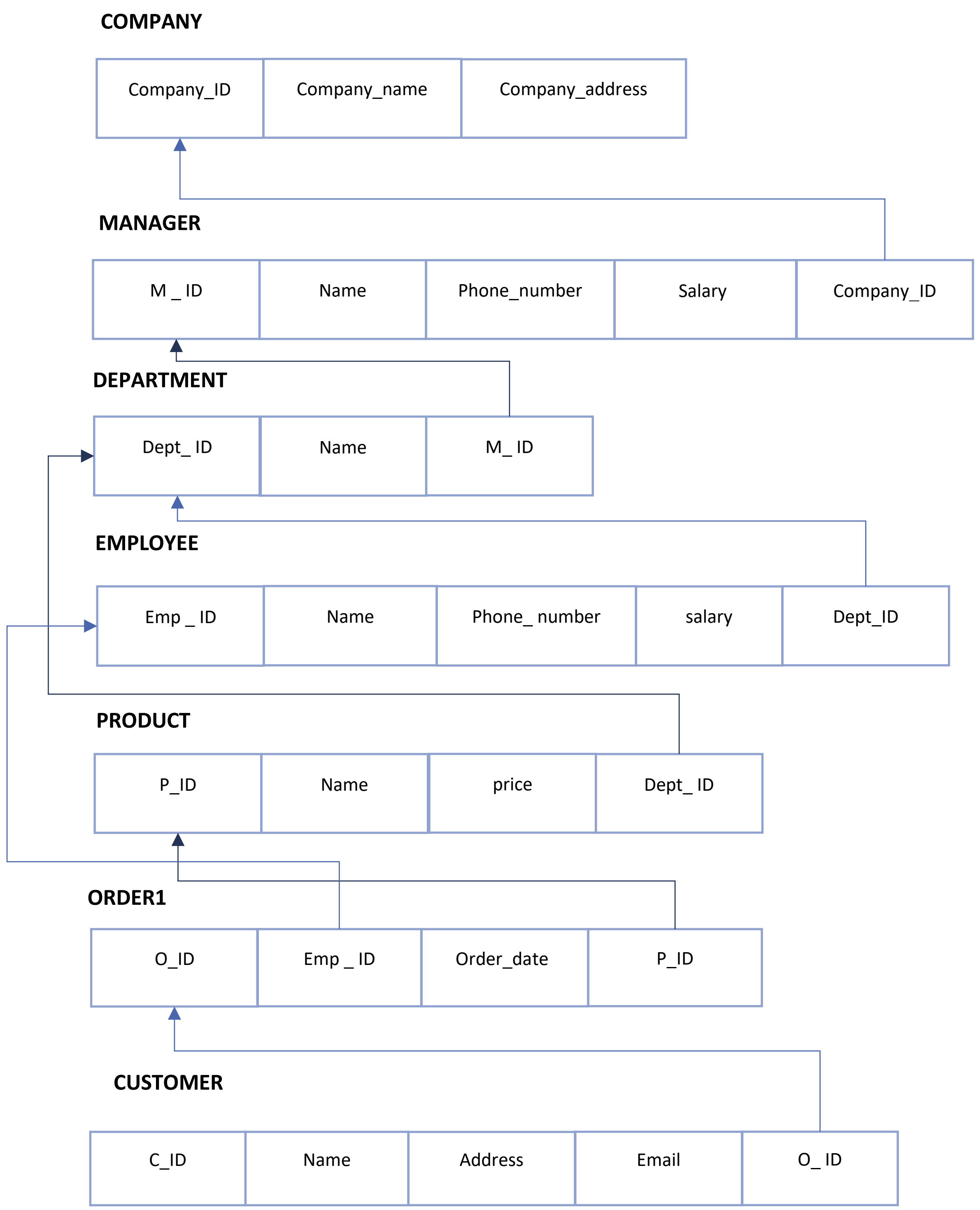
Emp_ID
N
N
Name
P_ID
Employe
Product
Name
Salary
Price
Dept_ID
N
N
Dept-ID
Phone_number
Work on
Included in
M
M
Order1
Order_date
O_ID
N
Emp_ID
P_ID
Belong to
1
Address
C_ID
Email
Customers
O_ID
Name

# PART 3: Logical Model:



# PART 4: Physical Data Model:

| Entity Type: COMPANY | | |
|---|---|---|
| **Attributes** | **Data Type** | **Required/Optional** |
| Company_ID | number(10) | Required |
| Company_name | VARCHAR(100) | Required |
| Company_address | VARCHAR(200) | Required |

| Entity Type: MANAGER | | |
|---|---|---|
| **Attributes** | **Data Type** | **Required/Optional** |
| M_ID | number(10) | Required |
| Name | VARCHAR(100) | Required |
| Phone_Number | VARCHAR(20) | Optional |
| Salary | number(10,2) | Required |
| Company_ID | number(10) | Required |

| Entity Type: DEPARTMENT | | |
|---|---|---|
| **Attributes** | **Data Type** | **Required/Optional** |
| Dept_ID | number(10) | Required |
| Name | VARCHAR(100) | Required |
| M_ID | number(10) | Required |

## Entity Type: EMPLOYEE

| Attributes | Data Type | Required/Optional |
|---|---|---|
| Emp_ID | NUMBER(10) | Required |
| Name | VARCHAR(100) | Required |
| Phone_Number | VARCHAR(20) | Optional |
| Salary | NUMBER(10,2) | Required |
| Dpet_ID | NUMBER(10) | Required |

## Entity Type: PRODUCT

| Attributes | Data Type | Required/Optional |
|---|---|---|
| P_ID | number(10) | Required |
| Name | VARCHAR(100) | Required |
| Price | number(10,2) | Required |
| Dpet_ID | NUMBER(10) | Required |

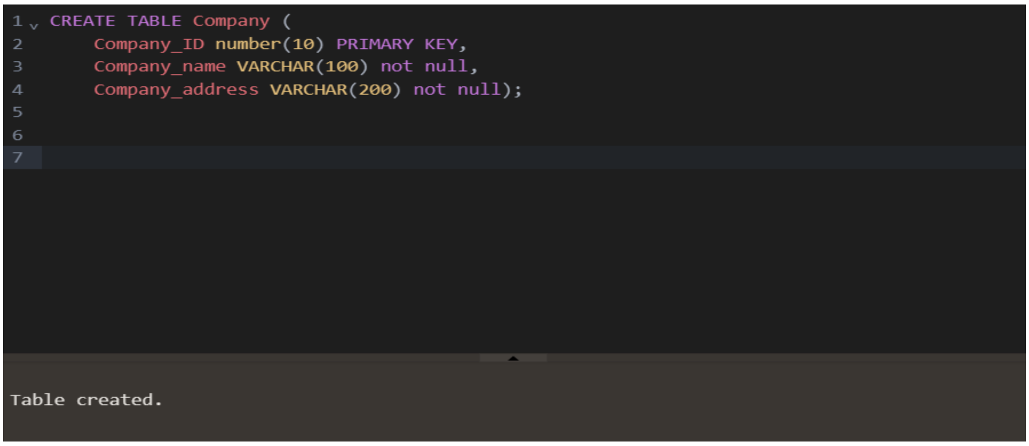
## Entity Type: ORDER1

| Attributes | Data Type | Required/Optional |
|---|---|---|
| O_ID | number(10) | Required |
| Emp_ID | number(١٠) | Required |
| P_ID | number(١٠) | Required |
| Order date | DATE | Required |

| Entity Type: CUSTOMR | | |
|---|---|---|
| **Attributes** | **Data Type** | **Required/Optional** |
| <u>C_ID</u> | number(10) | Required |
| Name | VARCHAR(100) | Required |
| Address | VARCHAR(112) | Required |
| Email | number(10) | Required |

# Creation of Database Table:

```
CREATE TABLE Company (
    Company_ID number(10) PRIMARY KEY,
    Company_name VARCHAR(100) not null,
    Company_address VARCHAR(200) not null);
```

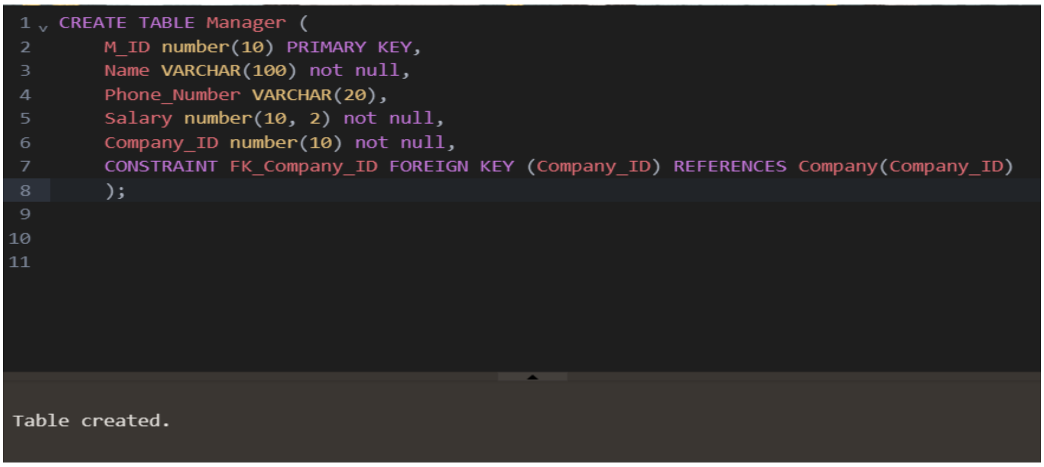


```
CREATE TABLE Company (
    Company_ID number(10) PRIMARY KEY,
    Company_name VARCHAR(100) not null,
    Company_address VARCHAR(200) not null);
```

```
Table created.
```

```
CREATE TABLE Manager (
    M_ID number(10) PRIMARY KEY,
    Name VARCHAR(100) not null,
```

```
    Phone_Number VARCHAR(20),
    Salary number(10, 2) not null,
    Company_ID number(10) not null,
    CONSTRAINT FK_Company_ID FOREIGN KEY (Company_ID) REFERENCES
Company(Company_ID)
    );
```




```
CREATE TABLE Department (
    Dept_ID number(10) PRIMARY KEY,
    Name VARCHAR(100) not null,
    M_ID number(10) not null,
    constraint FK_M_ID FOREIGN KEY (M_ID) REFERENCES Manager(M_ID)
    );
```

```
CREATE TABLE Employee (
    Emp_ID NUMBER(10) PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Phone_Number VARCHAR(20),
    Salary NUMBER(10, 2) NOT NULL,
    Dept_ID NUMBER(10) NOT NULL,
    CONSTRAINT FK_Dept_ID FOREIGN KEY (Dept_ID) REFERENCES
Department(Dept_ID));
```

```sql
 CREATE TABLE Product (
    P_ID number(10) PRIMARY KEY,
    Name VARCHAR(100) not null,
    Price number(10, 2) not null,
    Dept_ID NUMBER(10) NOT NULL,
    CONSTRAINT FK_Dept_ID2 FOREIGN KEY (Dept_ID) REFERENCES
Department(Dept_ID)
 );
CREATE TABLE Order1 (
    O_ID number(10) PRIMARY KEY,
    Emp_ID number(10) not null,
    P_ID number(10) not null,
    Order_Date DATE not null,
    CONSTRAINT FK_Emp_ID FOREIGN KEY (Emp_ID) REFERENCES
Employee(Emp_ID),
    CONSTRAINT FK_P_ID FOREIGN KEY (P_ID) REFERENCES Product(P_ID) );


CREATE TABLE Customer (
    C_ID number(10) PRIMARY KEY,
    Name VARCHAR(100) not null,
    Address varchar(112) not null,
    Email varchar(10) not null,
    O_ID number(10) ,
    CONSTRAINT FK_O_ID FOREIGN KEY (O_ID) REFERENCES Order1(O_ID) );
```

## Insertion Data:

**INSERT INTO Company VALUES (1, 'XYZ Corp', '456 Elm St');**

**INSERT INTO Company VALUES (2, 'ahmad ', '456 Elm St');**

```sql
INSERT INTO Company VALUES (3, 'VTEVIC', '754E CR');

INSERT INTO Manager VALUES (1, 'ali',  '053827616',10200.20, 1);
INSERT INTO Manager VALUES (2, 'omar', '054725468',10666.40, 2);
INSERT INTO Manager VALUES (3, 'Hasan','053756382',15600.40, 3);

INSERT INTO  Department VALUES(1,'fatemah',1);
INSERT INTO  Department VALUES(2,'amjad',2);
INSERT INTO  Department VALUES(3,'Atheer',2);

INSERT INTO  Employee VALUES(1,'ali','0538542675',1300.00,1);
INSERT INTO  Employee VALUES(2,'mohammed','0538576475',1300.00,2);
INSERT INTO  Employee VALUES(3,'sam','0538567843',1300.00,3);

INSERT INTO  Product VALUES(1, 'Laptop', 1000.99, 1);
INSERT INTO  Product VALUES (2, 'dress', 500.99, 2);
INSERT INTO  Product VALUES (3, 'Milk', 8.99, 3);

INSERT INTO  Order1 VALUES(1, 1, 1, '20-FEB-2023'),
INSERT INTO  Order1 VALUES (2, 2, 2, '23-FEB-2020'),
INSERT INTO  Order1 VALUES  (3, 3, 3, '21-FEB-2012');

INSERT INTO  Customer VALUES(1, 'MOhammed', '336 Elm St','m@mail.com',1);
INSERT INTO  Customer VALUES (2, 'Ahmad','456 Elm St' ,'a@mail.com',2);
INSERT INTO  Customer VALUES  (3, 'Ali', '885 Elm St' ,'l@mail.com',3);
```

## Some Queries to retrieve Data:

```
select * from Company;

select * from Employee;

select * from Department;

select * from Manager;

select * from product;

select * from Customer;

select * from Order1;
```